UML SEQUENCE DIAGRAM MODELLING

Online Grocery Store System

1. Aim

To model the main workflows of the Online Grocery Store System using UML sequence diagrams and explain the participants, messages, and interactions involved in each workflow.

2. Introduction

A UML sequence diagram is a behavioural diagram that shows how actors and system components communicate with one another in time order to complete a task. The participants are placed across the top, their lifelines extend vertically, and arrows show the messages exchanged. The diagram is read from top to bottom, so it explains both what happens and the order in which it happens.

3. Main Entities and Elements in the Diagrams

- **Customer (Actor):** The person using the grocery store application. The customer registers or logs in, browses products, adds items to the cart, places orders, makes payments, and checks order status.
- **Delivery Partner (Actor):** The person responsible for delivering an order. The delivery partner can log in and update the delivery status so that the customer can track progress.
- **Web/Mobile Application:** The user-facing part of the system. It displays forms and product information, collects customer input, sends requests to controllers, and shows success messages, errors, and order updates.
- **Auth Controller:** Handles registration and login requests. It checks whether an email already exists, creates user accounts, and validates login credentials.
- **Product Controller:** Processes product-related requests, such as retrieving the product list and product details, and checking whether the requested quantity is in stock.
- **Order Controller:** Manages checkout and order-related operations. It creates orders, retrieves order status, and processes delivery-status updates.
- **User Database:** Stores user information used during registration and login, such as account details and credentials needed for authentication.
- **Product Database:** Stores product information and availability data. The application requests product details and stock information through the relevant controller.
- **Order Database:** Stores order records and order-status information. It is updated when an order is created, payment status affects the order, or delivery progress changes.
- **Cart:** Temporarily holds the products selected by the customer before checkout. It is updated when an item is added and its quantity is accepted.
- **Payment Gateway:** Processes the payment request and returns a success or failure response. The system then updates the order accordingly.

4. UML Notation Used

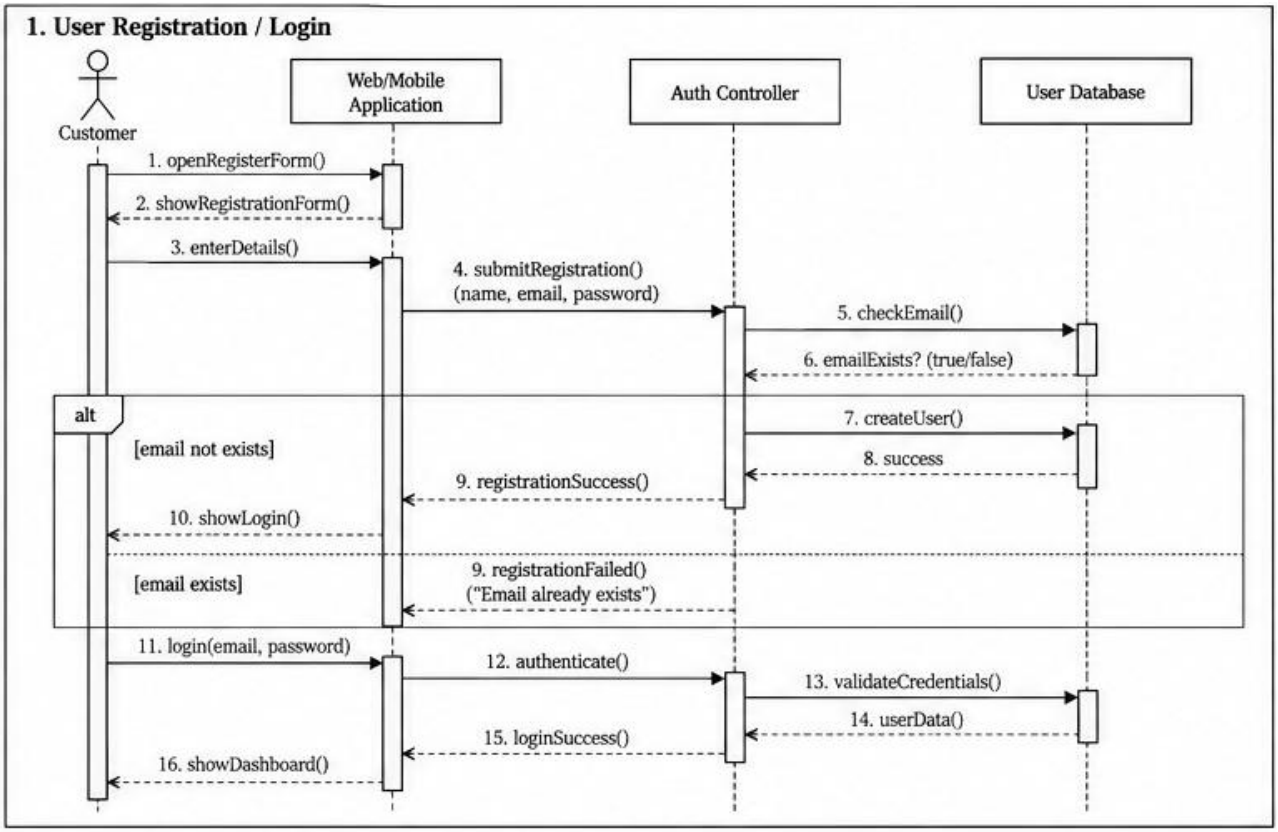
- **Actor:** A stick figure representing an external user or role that interacts with the system.
- **Lifeline:** A vertical dashed line below a participant's name, showing that participant during the interaction.
- **Message arrow:** A horizontal arrow showing a request, command, or response sent from one participant to another.

- **Activation bar:** A narrow rectangle on a lifeline showing when a participant is actively processing an operation.
- **alt fragment:** A framed area showing alternative outcomes based on a condition. For example, an email may already exist or not exist, stock may be available or unavailable, and payment may succeed or fail.
- **Return message:** A dashed arrow often used to show a response, such as a product list, authentication result, or saved order status.

5. Separated Sequence Diagrams and Workflow Explanation

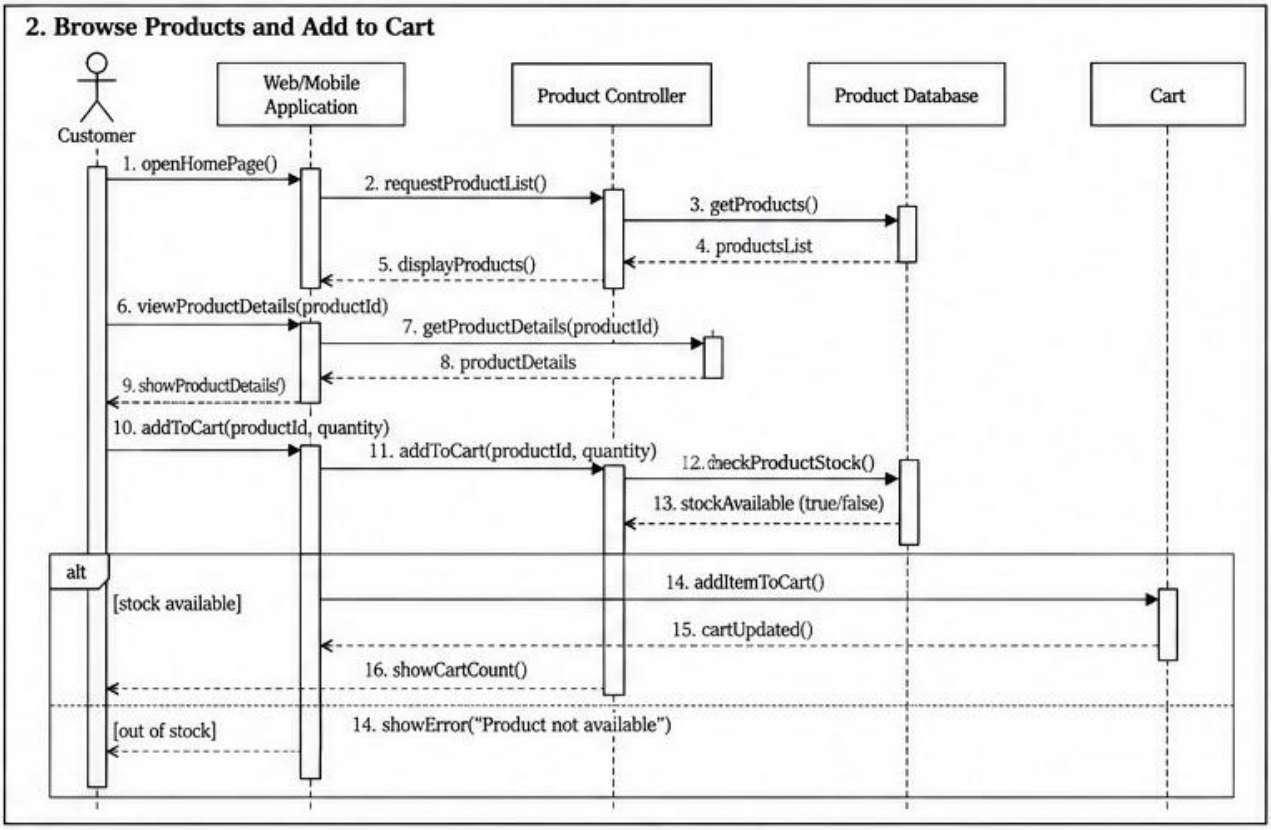
5.1 User Registration / Login

The customer opens the registration form and submits their name, email, and password. The Auth Controller checks the User Database to see whether the email already exists. If it is new, a user account is created; if it already exists, the application shows a registration error. During login, the controller validates the supplied credentials and displays the dashboard when authentication succeeds.



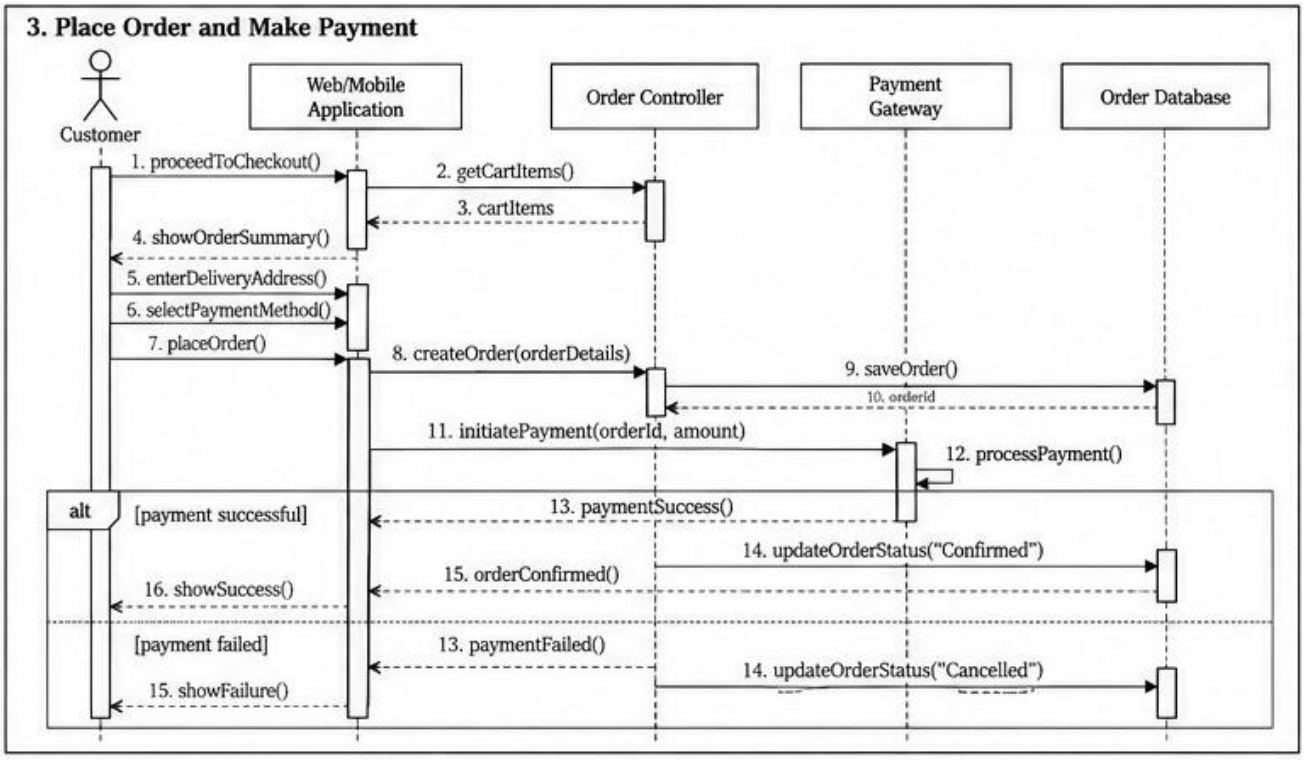
5.2 Browse Products and Add to Cart

The customer opens the home page and requests the product list. The Product Controller retrieves products from the Product Database and returns them to the application. The customer can view product details and add an item to the cart. The system checks stock availability: if stock is available, the Cart is updated; otherwise, an error message is displayed.



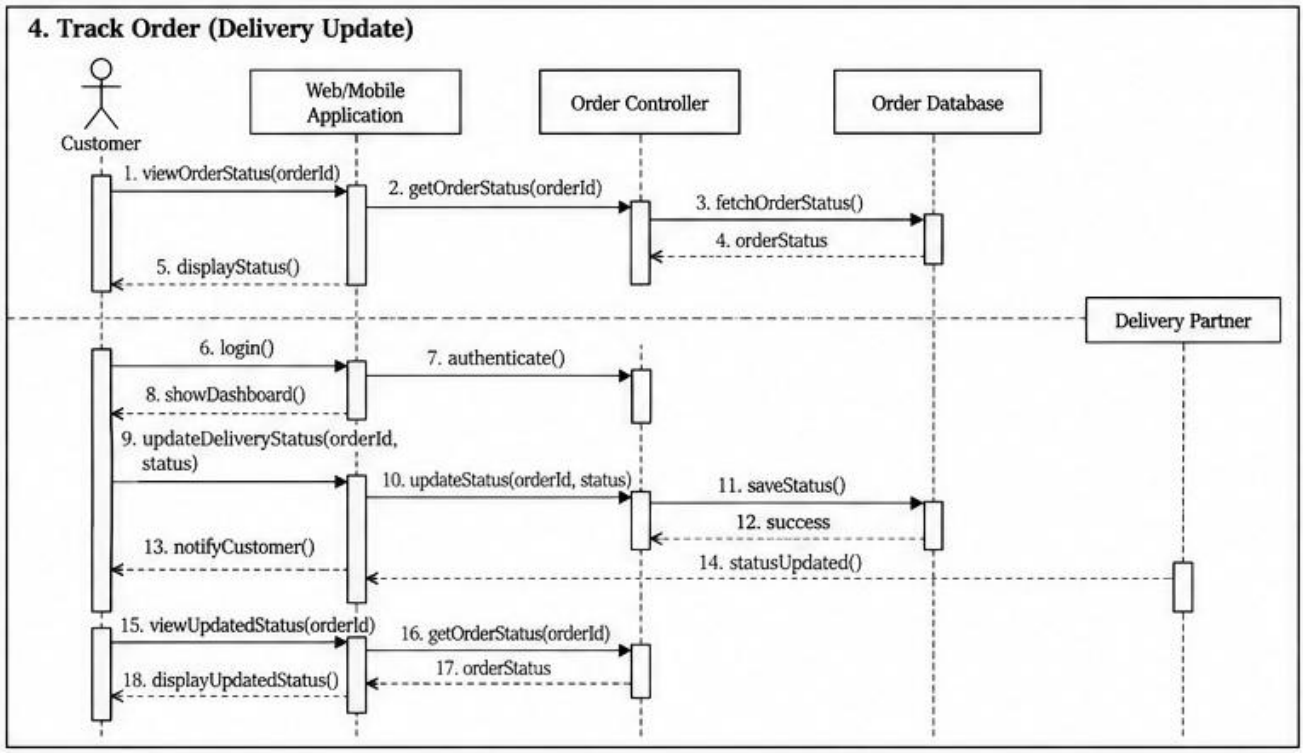
5.3 Place Order and Make Payment

The customer proceeds to checkout, reviews cart items, enters a delivery address, chooses a payment method, and places the order. The Order Controller saves the order in the Order Database and sends a payment request to the Payment Gateway. If payment succeeds, the order status becomes Confirmed; if payment fails, the order is marked as Cancelled and the application displays a failure message.



5.4 Track Order (Delivery Update)

The customer requests the current order status, which the Order Controller retrieves from the Order Database. The Delivery Partner can log in and submit a new delivery status. The system saves the update, notifies the customer, and allows the customer to view the latest order status.



6. Conclusion

The four sequence diagrams describe how the customer, delivery partner, application, controllers, databases, cart, and payment gateway interact during the main activities of the Online Grocery Store System. Separating the diagrams makes each workflow easier to understand and provides a useful reference for software design and implementation.

UML ACTIVITY DIAGRAM MODELLING

Online Grocery Store System

1. Aim

To model the main workflows of the Online Grocery Store System using UML activity diagrams, showing the actions, decisions, and responsibilities involved in each process.

2. Introduction

A UML activity diagram is a behavioural diagram that represents the flow of activities in a system. It shows the actions performed, the order in which they occur, and decision points that lead to different paths. It helps developers understand a process before implementing it in software.

3. Main Entities and Their Roles

- **Customer:** Registers or logs in, browses products, adds items to the cart, checks out, makes payments, and tracks orders.
- **System:** Validates details, displays pages and messages, checks order information, and coordinates with databases and payment services.
- **User Database:** Stores account information and supports credential checking, account creation, and loading the customer profile.
- **Product Database:** Stores product details and stock information used when customers browse and select products.
- **Cart:** Holds selected products and quantities and is updated when an item is successfully added.
- **Payment Gateway:** Processes payment requests and returns a success or failure result.
- **Order Database:** Stores order details and order status, including updates made during checkout and delivery.
- **Delivery Partner:** Views assigned orders and updates delivery progress, including marking an order as delivered.

4. UML Notations Used

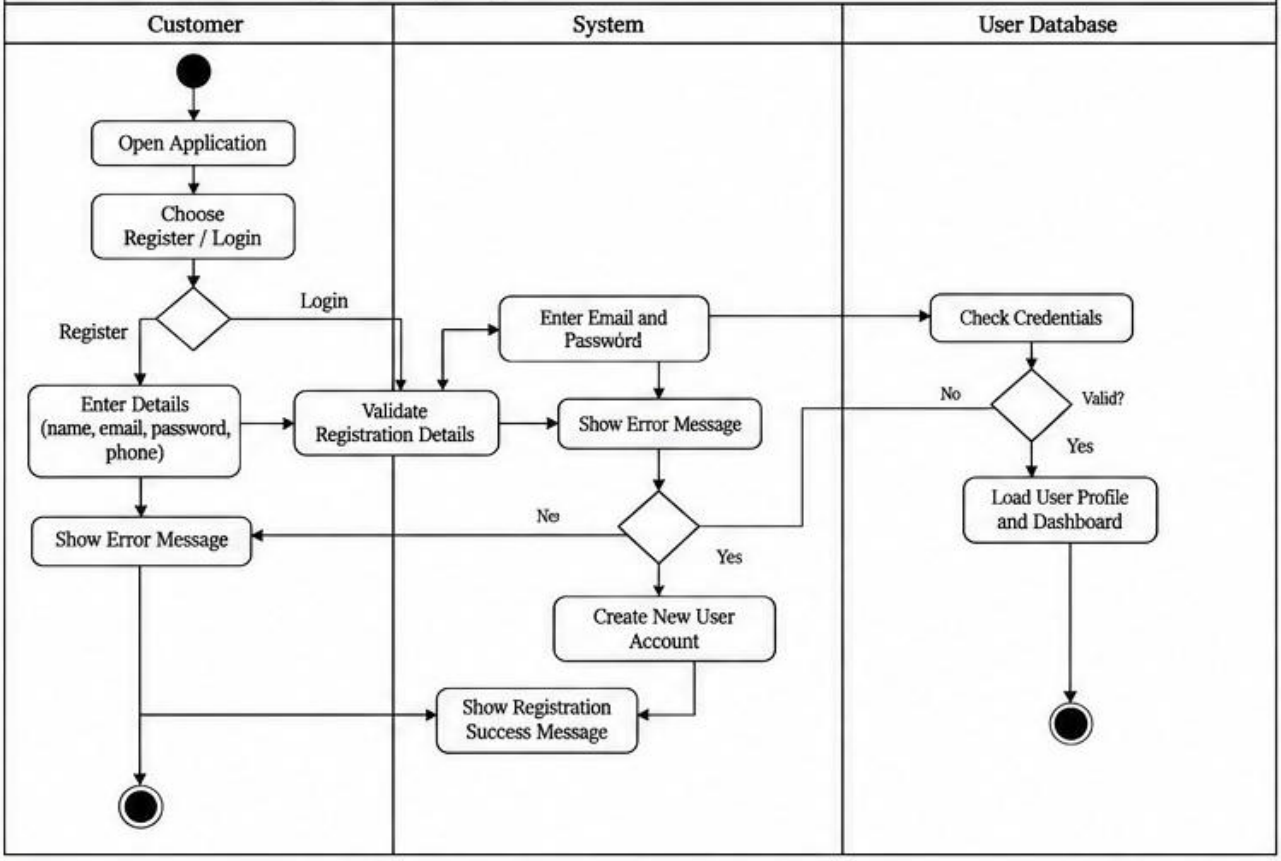
- **Initial node:** A filled black circle showing where the activity begins.
- **Action:** A rounded rectangle representing a task, such as entering details or validating an order.
- **Decision node:** A diamond where the flow branches based on a condition, such as valid/invalid details or stock availability.
- **Control flow:** An arrow showing the order in which activities are performed.
- **Swimlanes:** Sections labelled Customer, System, or Database that show responsibility for each activity.
- **Final node:** A bullseye-style circle indicating the end of the activity flow.

5. Activity Diagrams and Workflow Explanation

5.1 User Registration / Login Activity Diagram

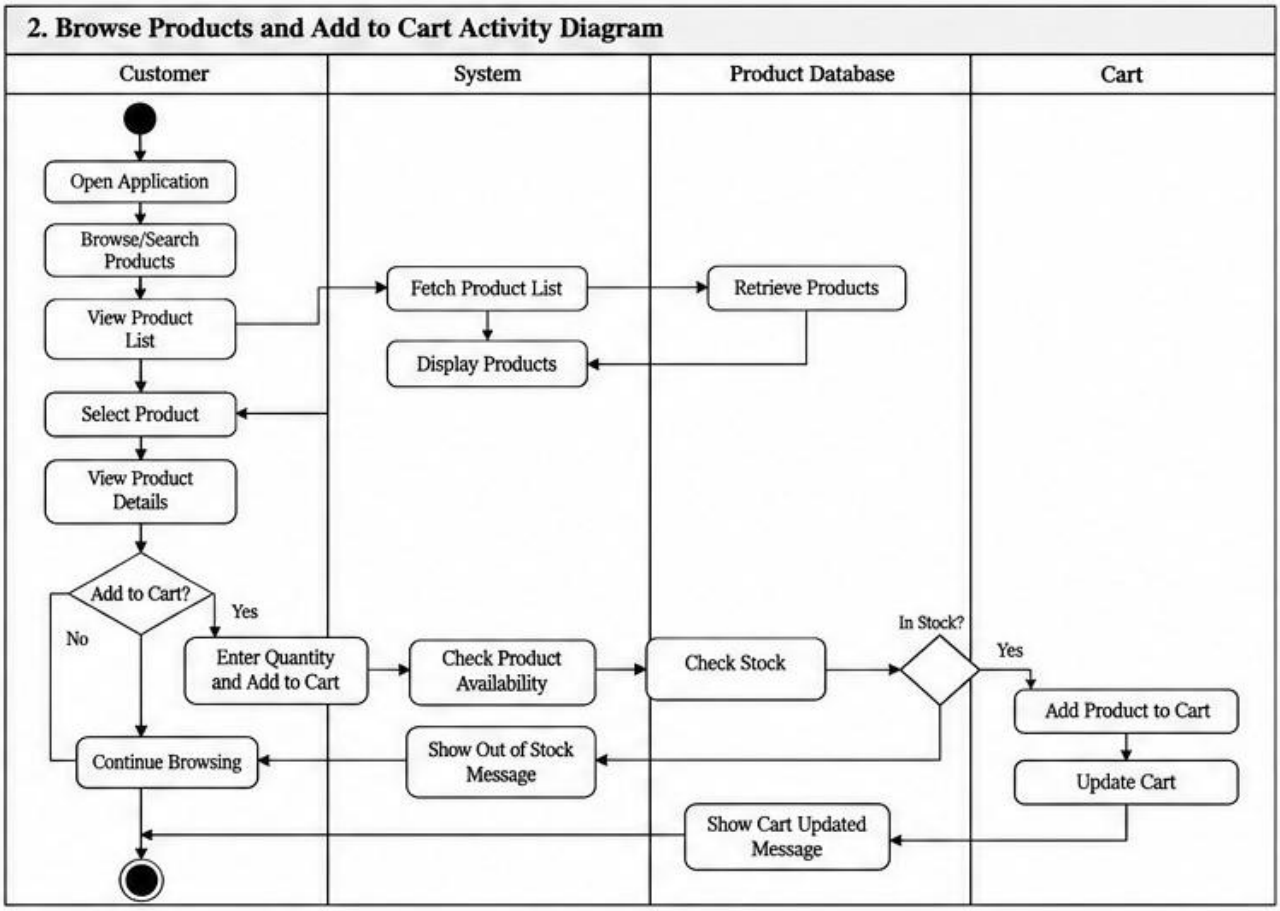
The customer opens the application and chooses registration or login. During registration, the system validates the details and checks the User Database. If the email is new and the details are valid, a new account is created; otherwise, an error message is shown. During login, the system checks the credentials and loads the profile and dashboard when they are valid.

1. User Registration / Login Activity Diagram



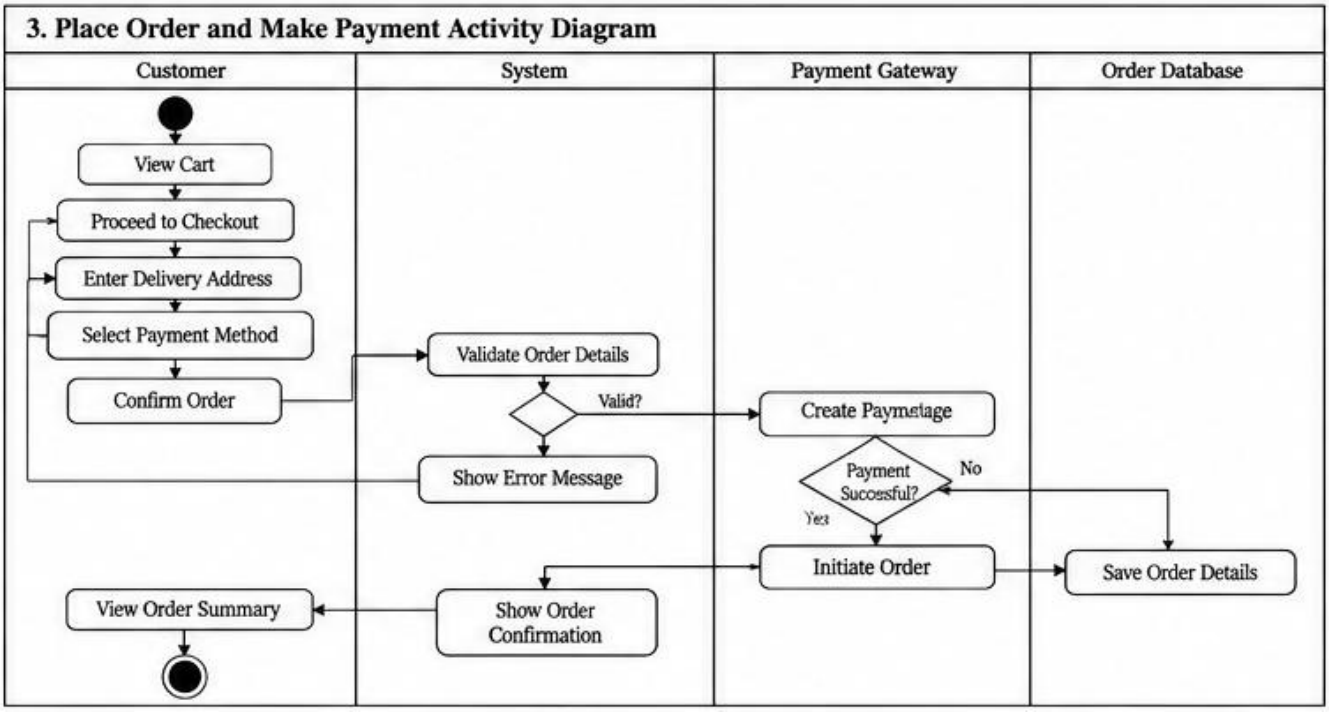
5.2 Browse Products and Add to Cart Activity Diagram

The customer browses products, views the product list, selects a product, and checks its details. If the customer chooses to add it to the cart, the system checks stock availability. If stock is available, the product is added and the cart is updated; otherwise, an out-of-stock message is displayed.



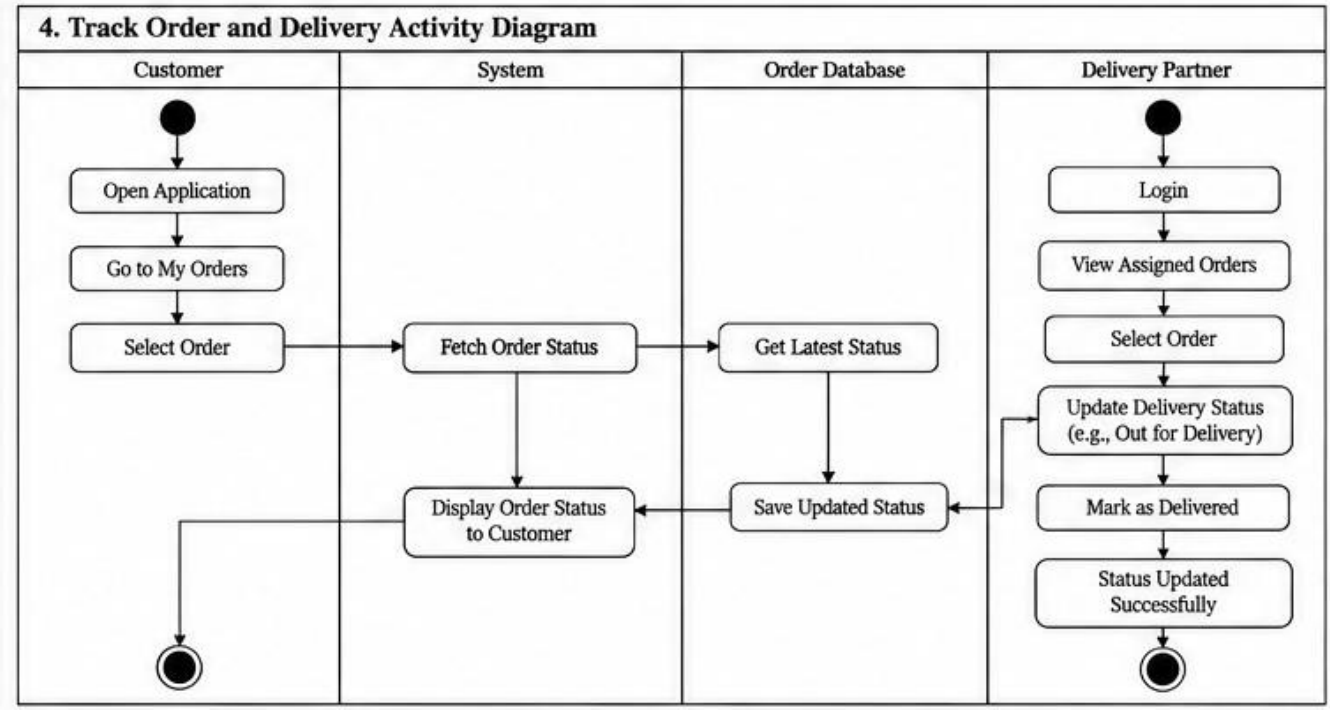
5.3 Place Order and Make Payment Activity Diagram

The customer views the cart, proceeds to checkout, enters a delivery address, selects a payment method, and confirms the order. The system validates the order details. If valid, the payment gateway processes the payment and order details are saved. If validation or payment fails, an error is shown; otherwise, order confirmation is displayed.



5.4 Track Order and Delivery Activity Diagram

The customer opens My Orders and selects an order to view its latest status. The system retrieves the status from the Order Database and displays it. The Delivery Partner logs in, views assigned orders, selects an order, updates its delivery status, and marks it as delivered when appropriate. The updated status is saved.



6. Conclusion

The activity diagrams describe the main processes of the Online Grocery Store System, from account access and product selection to checkout, payment, and delivery tracking. They show the actions, decisions, and responsibilities clearly and provide a useful reference for software design and implementation.